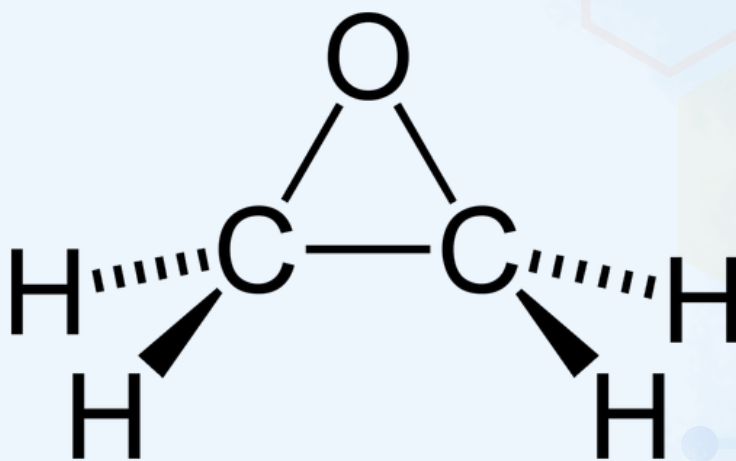


IUPAC name:
Oxirane

Molecular Formula:
C₂H₄O

Molar mass:
44.053 g/mol

Functional groups:
epoxide, cyclic ether,
three-membered ring



ETHYLENE OXIDE

“Carcinogenic Sterilizer”

Important Properties

- Sweet, ether-like organic compound
- **Water-soluble;** completely miscible in water
- **Melting point:** about -111 °C
- Contains a cyclic oxygen atom that affects polarity and ring strain
- Biologically toxic sterilizing agent

Uses/Value

- Used in medical sterilization, chemicals, and textiles
- Added to manufacturing lines for ethylene glycol and plastics
- Valuable commercial raw material in the chemical industry
- Provides high chemical reactivity for industrial synthesis

Structure & Identity

- **Common name:** Oxirane
- **Family:** Epoxide (or Cyclic ether)
- **Appearance:** Colorless gas/clear liquid under pressure
- **Natural sources:** Not naturally occurring; synthetically produced

Risks & Safety

- Highly toxic and carcinogenic; chronic exposure must be avoided to prevent cancer and reproductive hazards.
- Extremely flammable and explosive when exposed to air, heat, or catalytic surfaces.
- OSHA regulates strict airborne exposure limits, keeping the permissible limit to 1 ppm over an 8-hour workday.
- Exhibits a low boiling point of 10.4 °C, making it highly volatile and difficult to store safely at room temperature.

What Is Amazing About It?

Extremely reactive due to ring strain, so it easily breaks apart to bind with other molecules. Its structure comes from two carbon atoms and one oxygen atom: forming an oxirane ring.

Computer Engineering Relevance

Ethylene oxide production uses embedded systems to monitor critical temperature, pressure, and gas concentrations. Computer engineers develop the fail-safe sensors and automated control systems required to prevent thermal runaway and ensure safe manufacturing.

References:

- National Institute for Occupational Safety and Health. (2019). NIOSH pocket guide to chemical hazards (DHHS/NIOSH Publication No. 2005-149). U.S. Department of Health and Human Services, Centers for Disease Control and Prevention. <https://www.cdc.gov/niosh/docs/2005-149/>
- Occupational Safety and Health Administration. (1984). Ethylene oxide (29 CFR 1910.1047). U.S. Department of Labor. <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1047>
- Wikimedia Commons. (2021, June 3). Category:Ethylene oxide. https://commons.wikimedia.org/wiki/Category:Ethylene_oxide
- Wikipedia. (2026, September 6). Ethylene oxide. https://en.wikipedia.org/wiki/Ethylene_oxide